

Monomial insertions in the standard integral are a -derivatives of S_λ

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Abstract

The second §4.2 foothold: the one-dimensional mechanism behind the Taylor tree's $\partial^p S_\mu(\xi(0))$ terms. Expanding $e^{\beta\sqrt{n}u^k(\xi(u)-\xi(0))}$ inserts $(\sqrt{n}u^k)^p$ into the standard integral; we prove that in one dimension this insertion raises the order of the fluctuation function by $p/2$ while leaving $n^{-(h+1)/(2k)}$ untouched, and equals $\beta^{-p}\partial_a^p S_{(h+1)/(2k)}(a)$ by the general derivative ladder $\partial_a^p S_\lambda = \beta^p S_{\lambda+p/2}$ (also proved here for all $\lambda > 0$). Zero `sorry/axiom`.

1 Setting

The proof of `thm:TaylorTree` expands the standard integral into $\sum_p \frac{\beta^p}{p!} \int u^{h+m+n} (\sqrt{n}u^k)^p e^{-\beta n u^{2k} + \beta \sqrt{n} u^k \xi(0)} du$ and later rewrites each term through $\partial^p S_\mu$. In one dimension, with the unit-21 identity in hand, both steps are exact algebra: the insertion is a shift $h \mapsto h + kp$ absorbed by a $p/2$ shift of the fluctuation order, and the a -derivative ladder converts that shift back into a derivative.

2 The things formalised

```
theorem iteratedDeriv_fluctuation ( lam : ) (h : 0 < ) (hlam : 0 < lam) (p : ) :
  iteratedDeriv p (fun a => fluctuation lam a)
    = fun a =>   ^ p * fluctuation (lam + (p:)/2) a

theorem standardIntegral1D_insertion ( n a : ) (h k p : ) (hn : 0 < n) (hk : 0 < k) :
  ( u in Set.Ioi (0:), u^h * (Real.sqrt n * u^k)^p * Real.exp (...))
    = (1/(2*(k:))) * n^(-((h:)+1)/(2*k))
      * fluctuation (((h:)+1)/(2*k) + (p:)/2) a

theorem standardIntegral1D_insertion_deriv ... (h : 0 < ) ... :
  ( ... same integrand ...)
    = (1/(2*(k:))) * n^(-((h:)+1)/(2*k)) *   ^(-(p:))
      * iteratedDeriv p (fun a => fluctuation ((h:)+1)/(2*k)) a a
```

3 Proof strategy

The ladder is the same induction as `fluctuation_half_iteratedDeriv` with the order left symbolic: `iteratedDeriv_succ`, then `deriv_const_mul` and `deriv_fluctuation` at order $\lambda + p/2$, then the cast identity $\lambda + p/2 + 1/2 = \lambda + (p+1)/2$. The insertion identity rewrites the integrand as $(\sqrt{n})^p \cdot u^{h+kp} e^{\dots}$ (`mul_pow`, `pow_mul`, `pow_add`), pulls the constant through the integral, applies `standardIntegral1D_eq_fluctuation` at $h + kp$, and merges $(\sqrt{n})^p = n^{p/2}$ with $n^{-(h+kp+1)/(2k)}$ via `Real.rpow_add` into $n^{-(h+1)/(2k)}$. The derivative form is the ladder substituted, with $\beta^{-p}\beta^p = 1$.

4 Roadblocks and resolutions

None of substance — every step compiled on the first attempt. Two small idioms worth recording: $(\sqrt{n})^p = n^{p/2}$ is `Real.sqrt_eq_rpow` followed by `← Real.rpow_natCast` and `← Real.rpow_mul`; and the exponent identity $(h + kp + 1)/(2k) = (h + 1)/(2k) + p/2$ needs `push_cast` before `field_simp; ring` because $h + kp$ enters as a natural-number cast. Writing β^{-p} as a `zpow` `( ^(-(p:)))` keeps the statement honest about the sign and unfolds to $(\beta^p)^{-1}$ via `zpow_neg`, `zpow_natCast`.

5 What was Mathlib, what was new

Mathlib: `iteratedDeriv_succ`, `deriv_const_mul`, the `rpow` calculus, `integral_const_mul`. New: the general-order derivative ladder and the observation that the Taylor tree’s insertions-to-derivatives step is, in one dimension, a pure reindexing of the unit-21 identity.

6 Lessons

Once a change-of-variables identity is parametrised by the monomial exponent, every polynomial insertion is free; the paper’s “ $\partial^p S_\mu$ ” bookkeeping is the same fact read through the derivative ladder. Generalise the ladder to symbolic order as soon as it is needed at more than one value.

7 Follow-ups

The remaining §4.2 content is genuinely multivariate: the series interchange for $e^{\beta\sqrt{n}u^k(\xi(u)-\xi(0))}$ against the tree coefficients $\xi_{\mathbf{n},p}$, the bounded-domain remainder Δ , and the Mellin/meromorphic state-density machinery.